

Chart Descriptive Analysis

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Thursday 18th December, 2025



```
import geopandas as gpd
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

```
import sys
sys.path.append("..")
from CESTools.utils import clean_data
```

We sought to reproduce the charts of ‘Analysis of Race/Ethnicity and CalEnviroScreen 4.0 Scores’ report.

```
shape = gpd.read_file('../data/calenviroscreen40_shp/CES4 Final Shapefile.shp')
```

```
shape.columns
```

```
Index(['Tract', 'ZIP', 'County', 'ApproxLoc', 'TotPop19', 'CIScore',
      'CIScoreP', 'Ozone', 'OzoneP', 'PM2_5', 'PM2_5_P', 'DieselPM',
      'DieselPM_P', 'Pesticide', 'PesticideP', 'Tox_Rel', 'Tox_Rel_P',
      'Traffic', 'TrafficP', 'DrinkWat', 'DrinkWatP', 'Lead', 'Lead_P',
      'Cleanup', 'CleanupP', 'GWThreat', 'GWThreatP', 'HazWaste', 'HazWasteP',
      'ImpWatBod', 'ImpWatBodP', 'SolWaste', 'SolWasteP', 'PollBurd',
      'PolBurdSc', 'PolBurdP', 'Asthma', 'AsthmaP', 'LowBirtWt', 'LowBirWP',
      'Cardiovas', 'CardiovasP', 'Educate', 'EducateP', 'Ling_Isol',
      'Ling_IsolP', 'Poverty', 'PovertyP', 'Unempl', 'UnemplP', 'HousBurd',
      'HousBurdP', 'PopChar', 'PopCharSc', 'PopCharP', 'Child_10',
      'Pop_10_64', 'Elderly65', 'Hispanic', 'White', 'AfricanAm', 'NativeAm',
      'OtherMult', 'Shape_Leng', 'Shape_Area', 'AAPI', 'geometry'],
      dtype='object')
```

```
shape.head()
```

[illegible]

5 rows $\times$ 67 columns

```
df = clean_data(shape)
```

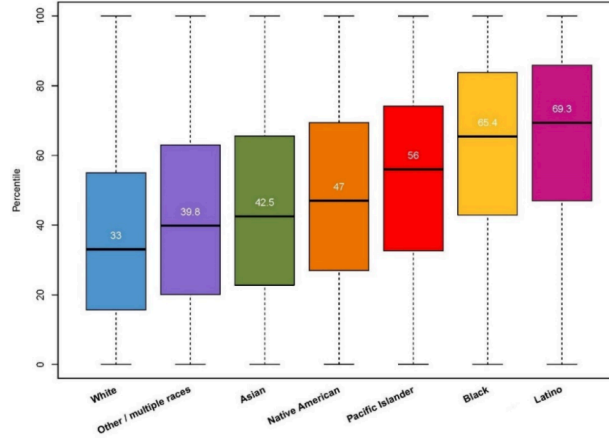
```
race_cols = ['Hispanic', 'White', 'AfricanAm', 'NativeAm', 'OtherMult', 'AAPI']
```

```
df['sum'] = df.loc[:, 'Hispanic'] + df.loc[:, 'White'] + df.loc[:, 'AfricanAm'] + df.loc[:,  
df[race_cols + ['sum']].describe()
```

	Hispanic	White	AfricanAm	NativeAm	OtherMult	AAPI	sum
count	8011.000000	8011.000000	8011.000000	8011.000000	8011.000000	8011.000000	8011.000000
mean	38.090437	38.671210	5.566892	0.418729	3.267521	13.985211	100.000000
std	26.434934	25.957581	8.575135	2.225242	2.445139	15.655214	0.000069
min	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	99.999800
25%	15.671250	14.726950	0.835100	0.000000	1.393800	3.234700	100.000000
50%	30.939400	37.246200	2.565900	0.000000	2.846400	8.429500	100.000000
75%	58.169800	60.991000	6.648450	0.360400	4.703750	18.505750	100.000000
max	100.000000	100.000000	84.708200	100.000000	107.065300	94.549900	100.000200

We verified that the race/ethnicity proportions within each county sum to 100%, which supports the reliability of the dataset.

Figure 1. CalEnviroScreen 4.0 Score by Racial or Ethnic Group



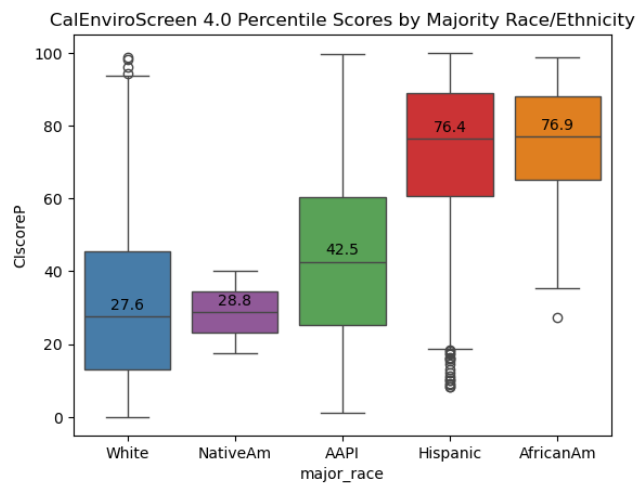
The original report generated this figure using individual-level data, where each observation represents a single Californian. With this structure, they were able to assign each person a CalEnviroScreen 4.0 percentile score based on their census tract and then group individuals by race/ethnicity to compute the accurate score distribution for each group.

However, our dataset is at the county level. While we had the total population data, generating the long-format dataset by repeatedly appending observations was avoided due to high time consumption. Thus we approximately

generated the boxplot by assigning each county to the race/ethnicity group with the highest proportion.

```
df.loc[:, 'major_race'] = df[race_cols].idxmax(axis=1)
median = df.groupby('major_race')['CIScoreP'].median().sort_values()
median_order = median.index
sns.boxplot(data= df, x='major_race', y='CIScoreP', order= median_order, palette="Set1", hue='major_race')
plt.title('CalEnviroScreen 4.0 Percentile Scores by Majority Race/Ethnicity')
for tick, label in enumerate(median_order):
    median_value = median[label]
    plt.text(tick, median_value+2, f'{median_value:.1f}', horizontalalignment='center')

plt.savefig("../visualizations/descriptive_charts/fig1_majorityrace_BaW.png", dpi=300, bbox_inches='tight')
/var/folders/0k/jn4h9ptx7_500msv80bg_rcm0000gn/T/ipykernel_9804/928823105.py:1: FutureWarning:
df.loc[:, 'major_race'] = df[race_cols].idxmax(axis=1)
```



```
df['major_race'].value_counts()
```

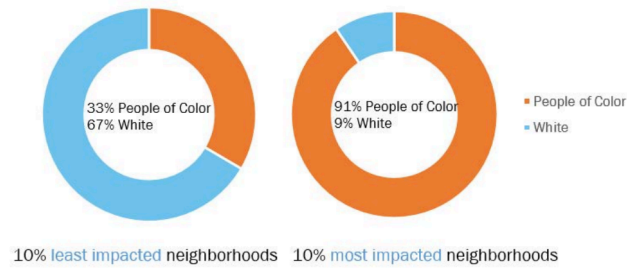
```
major_race
White      3853
Hispanic   3279
AAPI        763
AfricanAm   112
NativeAm     4
Name: count, dtype: int64
```

The comparison between the original figure 1 and the reproduced analysis confirms the severe structural environment injustice in California. The African American (corresponding to Black) and Hispanic (corresponding to Latino)

groups shows the consistently highest median vulnerability score, and the White group shows the lowest. This indicates that Hispanic and African American populations tend to live in communities with higher pollution burdens and vulnerabilities than the other/racial/ethnic groups analyzed.

However, the methodology used for the reproduction seemingly inflated the median scores significantly compared to the original report. Also, the extremely low count for the Native American group (n=4) makes its derived median unreliable for comparison purposes.

Figure 2. Race in the Least and Most Impacted Census Tracts of CalEnviroScreen 4.0.



```
bottom = df[df['CIScoreP'] <= 10]
top = df[df['CIScoreP'] >= 90]

bottom_white = bottom['White'].mean()
bottom_poc = 100 - bottom_white

top_white = top['White'].mean()
top_poc = 100 - top_white

labels = ['White', 'People of Color']
bottom_sizes = [bottom_white, bottom_poc]
top_sizes = [top_white, top_poc]

fig, ax = plt.subplots(1, 2, figsize = (10, 5))
ax[0].pie(bottom_sizes, labels=labels, autopct='%1.1f%%', startangle=90)
ax[0].set_title('10% least impacted neighborhoods')
ax[1].pie(top_sizes, labels=labels, autopct='%1.1f%%', startangle=90)
ax[1].set_title('10% most impacted neighborhoods')

plt.savefig("../visualizations/descriptive_charts/fig2_topbottom_piechart.png", dpi=300, bb
```

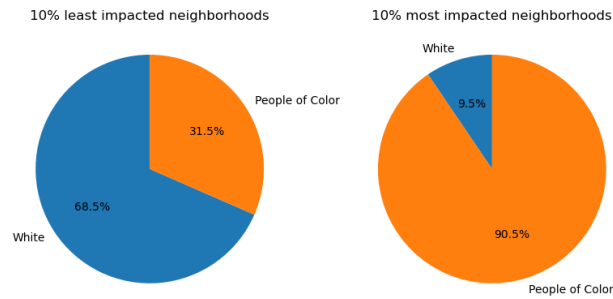
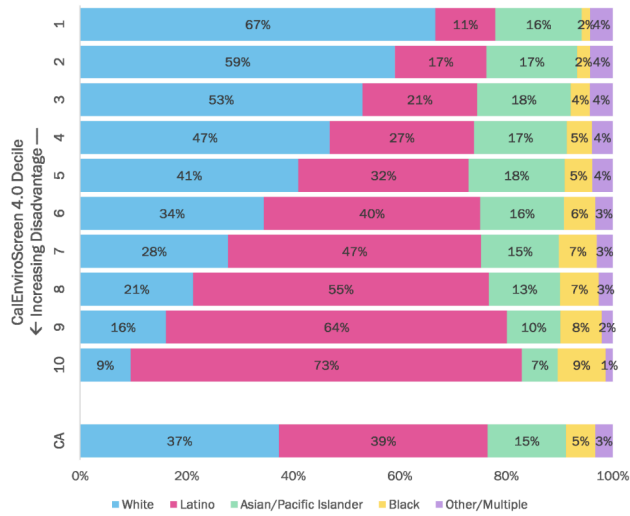


Figure 3. Racial Makeup of Each Decile of CalEnviroScreen 4.0 Score.

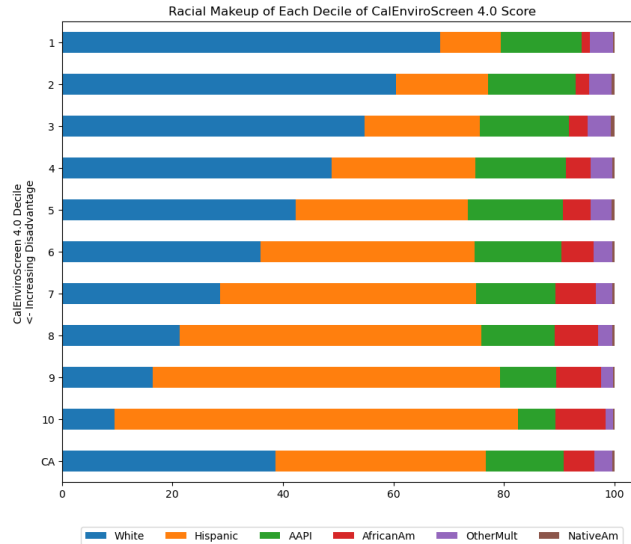


```
df['decile'] = pd.qcut(df['CIScoreP'], 10, labels=False) + 1
decile_means = df.groupby('decile')[race_cols].mean()
decile_means.index = list(range(1, 11))
ca_means = df[race_cols].mean()
ca_df = ca_means.to_frame().T
ca_df.index = ['CA']
df_plot = pd.concat([decile_means, ca_df], axis=0)
```

```
plot_order = ['White', 'Hispanic', 'AAPI', 'AfricanAm', 'OtherMult', 'NativeAm']
```

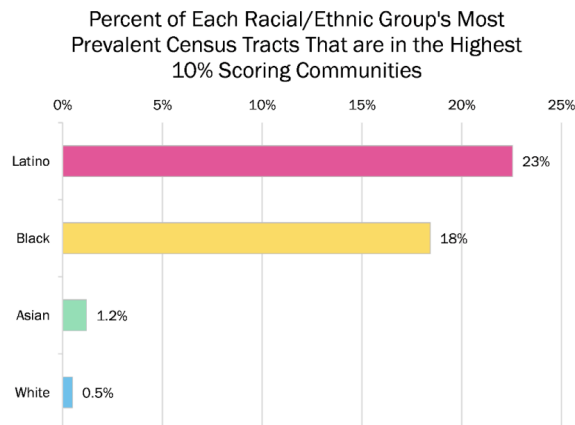
```
fig, ax = plt.subplots(figsize=(10, 8))
df_plot[plot_order].plot(kind='barh', stacked=True, ax=ax)
ax.set_ylabel('CalEnviroScreen 4.0 Decile \n< - Increasing Disadvantage')
ax.invert_yaxis()
ax.legend(loc='lower center', bbox_to_anchor = (0.5, -0.15), ncol = 6)
plt.title('Racial Makeup of Each Decile of CalEnviroScreen 4.0 Score')
```

```
plt.savefig("../visualizations/descriptive_charts/fig3_racialmakeupbydecile_barchart.png", c
```



The results are consistent with the original conclusions, reaffirming the environmental racial inequity within California. As environmental vulnerability increases (moving to higher deciles), the proportion of Hispanic and African American residents increases overwhelmingly. Conversely, the proportion of White residents decreases sharply as the disadvantage lessens, suggesting they predominantly reside in the most environmentally safe areas. This inequality is further highlighted when compared to the average bar representing the entire state of California at the bottom.

Figure 4. Most prevalent race by census tracts were overlaid with most disadvantaged 10% CalEnviroScreen scores.



```
top_counts = top['major_race'].value_counts()
```

```

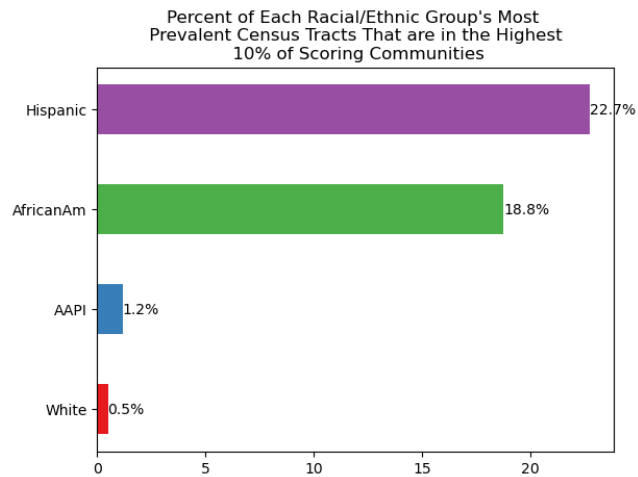
total_counts = df['major_race'].value_counts()
top_counts, total_counts

(major_race
Hispanic      745
AfricanAm     21
White         19
AAPI          9
Name: count, dtype: int64,
major_race
White      3853
Hispanic   3279
AAPI       763
AfricanAm  112
NativeAm    4
Name: count, dtype: int64)

top_ratio = (top_counts / total_counts).sort_values(ascending=True) * 100
plt.barh(top_ratio.index, top_ratio.values, height=0.5, color=sns.color_palette("Set1"))
for i, v in enumerate(top_ratio.values):
    plt.text(v, i, f'{v:.1f}%', va='center')
plt.title("Percent of Each Racial/Ethnic Group's Most \n Prevalent Census Tracts That are in")

plt.savefig("../visualizations/descriptive_charts/fig4_top10perctracts_barchart.png", dpi=300)

```



We were able to reproduce the exact same numerical values as the original report. Although the proportion of African Americans appears small in Figure 3, the fact that 18% of counties where African Americans are the major racial group are environmentally vulnerable (21 out of 112 counties) is not that different from the Hispanic proportion. This shows why attention should be paid not only to Hispanics but also to African Americans.

Additional Analysis

```
def select_high_correlated_vars(df, race_cols, threshold = 0.6):
    corr = df.drop(['major_race'], axis = 1).corr()
    vars = pd.DataFrame()
    for race_col in race_cols:
        race_corr = corr[race_col].drop(race_cols)
        selected = race_corr[abs(race_corr) > threshold].sort_values(ascending=False)
        vars = pd.concat([vars, selected], axis=1)
    return vars

select_high_correlated_vars(df, race_cols)
```

	Hispanic	White	AfricanAm	NativeAm	OtherMult	AAPI
EducatP	0.853965	- 0.775303	NaN	NaN	NaN	NaN
CIscoreP	0.741219	- 0.725284	NaN	NaN	NaN	NaN
decile	0.738075	- 0.720705	NaN	NaN	NaN	NaN
PopCharP	0.729622	- 0.713956	NaN	NaN	NaN	NaN
PovertyP	0.675304	NaN	NaN	NaN	NaN	NaN
Ling_IsolP	0.607499	- 0.753909	NaN	NaN	NaN	NaN

Excluding the CalEnviroScreen score, the correlation analysis shows that areas with a higher Hispanic population tend to have a larger proportion of residents with less than a high school education, higher poverty rates, and greater linguistic isolation, meaning more households with limited English proficiency. In contrast, areas with a higher White populations tend to have higher education levels and fewer linguistically isolated households.

```
bottom = df[df['EducatP'] <= 10]
top = df[df['EducatP'] >= 90]

bottom_white = bottom['White'].mean()
bottom_hispanic = bottom['Hispanic'].mean()
bottom_poc = 100 - bottom_white - bottom_hispanic

top_white = top['White'].mean()
top_hispanic = top['Hispanic'].mean()
top_poc = 100 - top_white - top_hispanic

labels = ['White', 'Hispanic', 'Other People of Color']
```

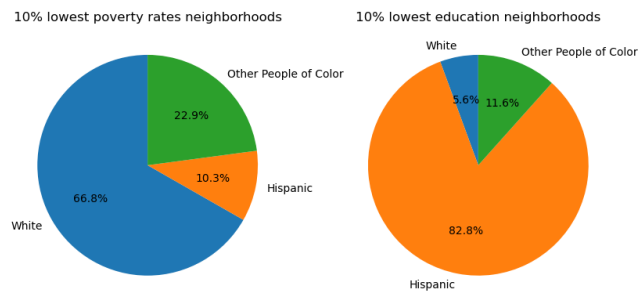
```

bottom_sizes = [bottom_white, bottom_hispanic, bottom_poc]
top_sizes = [top_white, top_hispanic, top_poc]

fig, ax = plt.subplots(1, 2, figsize = (10, 5))
ax[0].pie(bottom_sizes, labels=labels, autopct='%1.1f%%', startangle=90)
ax[0].set_title('10% lowest poverty rates neighborhoods')
ax[1].pie(top_sizes, labels=labels, autopct='%1.1f%%', startangle=90)
ax[1].set_title('10% lowest education neighborhoods')

plt.savefig("../visualizations/descriptive_charts/fig5_topbottom_educatP_piechart.png", dpi=

```



In the 10% lowest education neighborhoods, the Hispanic population accounts for an average of 82%. This concentration is notably stronger than in neighborhoods grouped by other variables.